We desire to avoid a messy integral

- We are close to being able to evaluate the desired equations:

$$f(x_k \mid \mathbb{Z}_{k-1}) = \int_{R_{x_{k-1}}} f(x_k \mid x_{k-1}) f(x_{k-1} \mid \mathbb{Z}_{k-1}) \, \mathrm{d}x_{k-1}$$

$$f(x_k \mid \mathbb{Z}_k) = \frac{f(z_k \mid x_k) f(x_k \mid \mathbb{Z}_{k-1})}{f(z_k \mid \mathbb{Z}_{k-1})}.$$

- Typically, we do not calculate the denominator term $f(z_k \mid \mathbb{Z}_{k-1})$ because:
  - It involves evaluation of an integral (via law of total probability).
  - It is a scalar value, independent of the value of x_k.
  - It serves only to normalize $f(x_k \mid \mathbb{Z}_k)$ so that $\int_{R_{x_k}} f(x_k \mid \mathbb{Z}_k) \, \mathrm{d}x_k = 1$.
- Thus, we will choose to evaluate only the numerator of $f(x_k \mid \mathbb{Z}_k)$—which is proportional to $f(x_k \mid \mathbb{Z}_k)$—but not the pdf $f(x_k \mid \mathbb{Z}_k)$ itself.
  - But, how does doing this affect Monte–Carlo integration?

Unnormalized $f(x_k \mid \mathbb{Z}_k)$ biases prediction integral

- Let's consider what happens to the prediction integral:

$$f(x_k \mid \mathbb{Z}_{k-1}) = \int_{R_{x_{k-1}}} f(x_k \mid x_{k-1}) f(x_{k-1} \mid \mathbb{Z}_{k-1}) \, \mathrm{d}x_{k-1}$$

  if we do not normalize $f(x_{k-1} \mid \mathbb{Z}_{k-1})$ correctly to have volume of 1.
- By now, we have rewritten this prediction integral as:

$$\mu = \int_{R_x} g(x) \left[\frac{f(x)}{q(x)} \right] q(x) \, \mathrm{d}x \approx \frac{1}{N} \sum_{i=1}^{N} g(x^{(i)}) \left[\frac{f(x^{(i)})}{q(x^{(i)})} \right] = \frac{1}{N} \sum_{i=1}^{N} g(x^{(i)}) w(x^{(i)}),$$

  where $f(x)$ integrates to 1. What if we substitute $\tilde{f}(x)$, which doesn't?
- Since $\tilde{f}(x)$ is unnormalized, $\tilde{f}(x) = c\, f(x)$, where c is an unknown constant:

$$\hat{\mu}_b = \int_{R_x} g(x) \left[\frac{\tilde{f}(x)}{q(x)} \right] q(x) \, \mathrm{d}x \approx \frac{1}{N} \sum_{i=1}^{N} g(x^{(i)}) \left[\frac{\tilde{f}(x^{(i)})}{q(x^{(i)})} \right] = \frac{c}{N} \sum_{i=1}^{N} g(x^{(i)}) \left[\frac{f(x^{(i)})}{q(x^{(i)})} \right] = c\hat{\mu}.$$

Computing the bias

- So, if we do not normalize $f(x_k \mid \mathbb{Z}_k)$, our calculation of the prediction integral is biased.
  - Instead of computing $f(x_k \mid \mathbb{Z}_{k-1})$, we compute $c f(x_k \mid \mathbb{Z}_{k-1})$.
- We can compute the bias in the Monte–Carlo calculation if we replace the arbitrary function $g(x)$ by 1 (conceptually); i.e., we seek to find $\mathbb{E}[1] = 1$ to expose the bias.

$$\hat{\mu}_1 = \int 1 \cdot \left[\frac{\tilde{f}(x)}{q(x)} \right] q(x) \, \mathrm{d}x \approx \frac{1}{N} \sum_{i=1}^{N} 1 \cdot \left[\frac{\tilde{f}(x^{(i)})}{q(x^{(i)})} \right] = c \underbrace{\frac{1}{N} \sum_{i=1}^{N} \frac{f(x^{(i)})}{q(x^{(i)})}}_{1} = c.$$

- If we equate the second form of the equation to the final result,

$$c = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{\tilde{f}(x^{(i)})}{q(x^{(i)})} \right] = \frac{1}{N} \sum_{i=1}^{N} \widetilde{w}(x^{(i)}).$$

- If we divide our biased estimate of $f(x_k \mid \mathbb{Z}_{k-1})$ by c, we remove the bias.

Removing the bias

- So, for any general $g(x)$, we can compute unbiased:

$$\begin{aligned}\hat{\mu}_N &= \frac{1}{N}\sum_{i=1}^{N} g(x^{(i)})\left[\frac{f(x^{(i)})}{q(x^{(i)})}\right] = \frac{1}{N}\sum_{i=1}^{N} g(x^{(i)})w(x^{(i)}) \\ &= \frac{1}{N}\sum_{i=1}^{N} g(x^{(i)})\left[\frac{\tilde{f}(x^{(i)})}{q(x^{(i)})}\right]\times\frac{1}{c} = \frac{1}{N}\sum_{i=1}^{N} g(x^{(i)})\left(\frac{\widetilde{w}(x^{(i)})}{\frac{1}{N}\sum_{j=1}^{N}\widetilde{w}(x^{(j)})}\right) \\ &= \sum_{i=1}^{N} g(x^{(i)})w^*(x^{(i)}),\end{aligned}$$

  where $w^*(x^{(i)}) = \frac{\widetilde{w}(x^{(i)})}{\sum_{j=1}^{N}\widetilde{w}(x^{(j)})}$. Notice that the $1/N$ factors cancel.
- This last step is called weight normalization.

Applying to the prediction step

- Instead of computing: $f(x_k \mid \mathbb{Z}_k) = \frac{f(z_k|x_k)f(x_k|\mathbb{Z}_{k-1})}{f(z_k|\mathbb{Z}_{k-1})}$,
  we will compute: $\tilde{f}(x_k \mid \mathbb{Z}_k) = f(z_k \mid x_k)f(x_k \mid \mathbb{Z}_{k-1})$.
- Then, the prediction integral:

$$\begin{aligned}f(x_k \mid \mathbb{Z}_{k-1}) &= \int_{R_{x_{k-1}}} f(x_k \mid x_{k-1})f(x_{k-1} \mid \mathbb{Z}_{k-1})\,\mathrm{d}x_{k-1} \\ &\approx \frac{1}{N}\sum_{i=1}^{N} f(x_k \mid x^{(i)})\left[\frac{f(x^{(i)} \mid \mathbb{Z}_{k-1})}{q(x^{(i)})}\right] = \frac{1}{N}\sum_{i=1}^{N} f(x_k \mid x^{(i)})w(x^{(i)}) \\ &= \sum_{i=1}^{N} f(x_k \mid x^{(i)})\frac{\left[\frac{\tilde{f}(x^{(i)}|\mathbb{Z}_{k-1})}{q(x^{(i)})}\right]}{\left[\sum_{j=1}^{N}\frac{\tilde{f}(x^{(j)}|\mathbb{Z}_{k-1})}{q(x^{(j)})}\right]} = \sum_{i=1}^{N} f(x_k \mid x^{(i)})w^*(x^{(i)}).\end{aligned}$$

- We no longer need to normalize $f(x_k \mid \mathbb{Z}_k)$ to have unit volume!

Prior example reconsidered

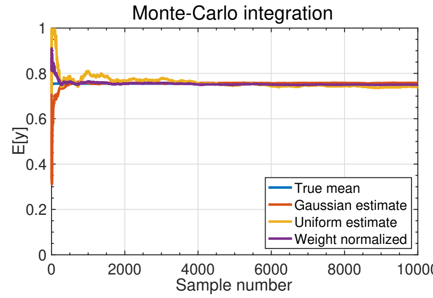
- Consider the same example as last lessons: we desire to find $\mathbb{E}[y]$, where $y = \cos(x)$ and $x \sim \mathcal{N}(0, \sigma_x^2)$.
- Again, we use an importance density, $x \sim \mathcal{U}(-R, R)$ for which $f(x) = \frac{1}{2R}$.
- We re-write the problem as: $\mathbb{E}[y] = \int_{-\infty}^{\infty} \frac{\cos(x)\frac{1}{\sqrt{2\pi\sigma_x^2}}\exp\left(\frac{-x^2}{2\sigma_x^2}\right)}{\frac{1}{2R}}\left(\frac{1}{2R}\right)\mathrm{d}x$.
- Comparing to our integral form, $\mu = \int [g(x)f(x)/q(x)]\,q(x)\,\mathrm{d}x$, we again assign:
  $g(x) = \cos(x)$, $\quad f(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}}\exp\left(\frac{-x^2}{2\sigma_x^2}\right)$, and $q(x) = \frac{1}{2R}$.
- But, now let $\tilde{f}(x) = \exp\left(\frac{-x^2}{2\sigma_x^2}\right)$ and $\widetilde{w}(x) = \tilde{f}(x)/q(x) = 2R\exp\left(\frac{-x^2}{2\sigma_x^2}\right)$.
- So, we will draw N independent uniform RVs $x^{(i)}$ between $-R$ and R and compute:

$$\mathbb{E}[y] \approx \sum_{i=1}^{N}\cos(x^{(i)})\widetilde{w}(x^{(i)}) \Big/ \sum_{j=1}^{N}\widetilde{w}(x^{(j)}).$$

Octave code for example

```
N = 10000; sx = 0.75;  % problem constants
q = 8*sx*(rand([1 N])-0.5); % generate unif RVs
wtilde = exp(-p.^2/(2*sx^2)) * 8*sx; % R = 4*sx
mu = cumsum(cos(p).*wtilde)./cumsum(wtilde); % running approx
plot(1:N,mu);            % plot estimate versus N
```

- I chose $q(x) \sim \mathcal{U}(-4\sigma_x, 4\sigma_x)$. Does not span entire support of $f(x)$, but "close enough".
  - Convergence turns out to be faster than before.
  - However, we get the same value in the end.
  - Results will (again) be somewhat different from one trial run to another due to the randomness of the Monte–Carlo method.



Summary

- We desire to make the particle filter as computationally simple as possible, without sacrificing accuracy.
- We have noticed that the only purpose for computing $f(z_k \mid \mathbb{Z}_{k-1})$—which requires a messy integral—is to normalize the pdf $f(x_k \mid \mathbb{Z}_k)$.
- You learned that if we choose *not* to normalize $f(x_k \mid \mathbb{Z}_k)$, we can compensate for this omission by modifying the prediction step.
- Instead of using normalized weights $w(x^{(i)})$ when evaluating the prediction step, we use unnormalized weights $\widetilde{w}(x^{(i)})$ and then perform weight normalization to correct the final calculation.
- Even if x_k is multi-dimensional, weight-normalization requires only a 1-d summation, so is fast.